

The denominator used for calculating maternal mortality should be specified as either the number of live births or the number of total births (live births plus fetal deaths). Where both denominators are available, a calculation should be published for each.

Results should be expressed as a ratio of the numerator over the denominator, multiplied by k (where k may be 1000, 10,000 or 100,000, as preferred and indicated by the country). Maternal mortality ratios and rates can thus be expressed as follows:

1. Maternal mortality ratio: (Maternal deaths/Live births or total births) x k
2. Direct obstetric mortality ratio: (Direct obstetric death only/Live births or total births) x k
3. Ratio for death occurring during pregnancy, childbirth and puerperium: (Deaths occurring during pregnancy, childbirth and puerperium/Live births) x k

3 Part 3 - New in ICD-11

3.1 ICD-11 new conventions and terminology

Table 1: *Major changes from ICD-10 to ICD-11, including rationale*

ICD-10

Coding Scheme

Chapter numbering is roman numerals
Three-character categories, each of which can be further divided into up to 10 four-character subcategories.

Alphanumeric code with a letter in the first position and a number in the second, third and fourth positions. The fourth character follows a decimal point.

The first character of a code is a letter and does not relate to the chapter number. The letter may have been the same for two short chapters (e.g. Chapter VII (H00-H5) and Chapter VIII (H60-H95)), or two letters may have been used for one long chapter (e.g. Chapter XIX (S00-T98)).

Residual category identified by numeric character .8 and unspecified category identified by numeric character .9.

Code cluster concept does not exist in ICD-10.

Terminology

a range of expressions are used to describe a causal relationship between conditions in a code title

ICD-11

Chapter numbering is Arabic

Stem code (category) is four characters and there are two levels of subcategories

An alphanumeric code with a letter in the second position and number in the third character position to differentiate from the codes of ICD-10. The inclusion of a forced number at the third character position prevents spelling 'undesirable words'. A letter in the second character position allows for clear distinction between a code from ICD-11 and one from ICD-10. Alphanumeric codes cover the range from 1A00.00 to ZZ9Z.ZZ. Codes starting with an 'X' indicate an extension code (see Extension code chapter). The letters 'O' and 'I' are omitted to prevent confusion with the numbers '0' and '1'.

The first character of the code always relates to the chapter. A first character of 1-9 is used for chapters 01 through 09 and for chapters 10 through 26, the first character is a letter. The code range of a single chapter always has the same character in the first position. For example, [1A00](#) is a code in chapter 01, and [BA00](#) is a code in chapter 11.

The terminal letter 'Y' is reserved for the residual category 'other specified' and the terminal letter 'Z' is reserved for the residual category 'unspecified'.

ICD-11 supports postcoordination and the linking codes within a code cluster.

The preferred term is 'due to' for categories where two conditions are mentioned and causal sequence exists. Other terms, such as 'caused by'; or 'attributed to' may be allowed synonyms. The phrase 'secondary to' is equivalent and may also be included as a synonym.

ICD-10

a range of expressions indicating the concurrence of two conditions in a code title (e.g. 'in' or 'with').

ICD-11

The preferred term is 'associated with' for categories where two conditions are mentioned and there is no causal sequence implied.

Dagger-Asterisk system and additional sub-classifications

	ICD-10	ICD-11
Dagger asterisk system	ICD-10 (and ICD-9) used the dagger asterisk system to describe the aetiological condition for primary tabulation (dagger code) and the clinical manifestation, relevant site and or other aspects (asterisk code). In addition, there were sets of codes to be used to add more detail (e.g. B95-B97) or lists of sub-classifications to add anatomical detail to categories.	The ICD-11 equivalents of asterisk codes (i.e. codes for manifestations) and other codes that served to add detail, may be found in Chapter 21 'Symptoms, signs or clinical findings, not elsewhere classified', the Chapter X 'Extension codes' chapter or in a body system chapter as appropriate. The extension codes include chapter groups anatomy, agents, histopathology and other aspects that may be used to add detail to a code.
Use of multiple codes for one condition/additional sub-classifications	More than one category could be used to specify more detail for another category. For example, infectious agents (B95-B97) or the asterisk codes.	Postcoordination - The use of multiple codes (i.e. stem codes and/or extension codes) together to fully describe a documented clinical concept.
'Code also' instruction'	'Use additional code, if desired, to identify' notes were present to suggest optional coding.	'Code also' instructions inform about additional information that needs to be coded in conjunction with certain categories, because that additional information is relevant for primary tabulation

The dagger and asterisk system has been removed in ICD-11, but the functionality of coding the aetiology and manifestation remains. A number of former asterisk codes that were previously used to identify manifestations of diseases are now listed in Chapter 21 'Symptoms, signs, or clinical findings, not elsewhere classified'. A portion of former asterisk codes also reside in the corresponding body system chapter. Asterisk codes that were repetitions of the dagger code were removed. Lists for coding optional anatomical detail have been grouped into one section in Chapter X 'Extension codes'.

Other general differences

	ICD-11
Category description	All ICD-11 categories have a short and a long description. The short description describes the meaning of the category in 100 words or less and appears in the printed version of the classification. The long description is without length restriction, including detailed information that appears in the content model.
Content model	All ICD-11 categories include separate information on anatomy, aetiology and other aspects that can be accessed for search purposes, or when browsing in the tabular list of the ICD-11 MMS.

Special tabulation lists of ICD-10 continue to exist, but there are two additional ones, the Startup Mortality List (SMoL) and the list for verbal autopsy. Additional special tabulations can be derived from the new multiple parenting technique, e.g. all WHO notifiable diseases, listing all conditions that are assigned to the relevant section of the infectious diseases chapter.

For morbidity, the definition of main diagnosis has changed to be ‘the reason for admission, after assessment at the end of the stay’. This definition is less prone to interpretation, and countries that had switched from the ‘most resource intensive’ definition to the ‘reason for admission at the end of the stay’ using ICD-10, noticed only small changes in their activity statistics.

3.1.1 Short Description

The description is a short characterisation (maximum of 100 words) of the entity that states things that are always true about a disease or condition and necessary to understand the scope of the rubric. Descriptions do not contain elements intended for use in level 3 (common epidemiology) or things that may be true for level 4 (clinical criteria). Descriptions were formerly called ‘short definitions’. For further information refer to section [3.4The Content Model](#)

3.1.2 Additional Information

This is a text field that is not mandatory, but that may contain any additional information about, or characteristics of, the diseases or conditions included in the entity. This text field provides more context for the entity. For example, the most common epidemiologic circumstances, putative or highly suspected aetiologic agents, or other information that may not always be true but may be common, typical, or expected. Additional information was formally called ‘long definition’.

3.1.3 Code Structure

The codes of the ICD-11 are alphanumeric and cover the range from 1A00.00 to ZZ9Z.ZZ. Codes starting with ‘X’ indicate an extension code (see Extension codes). The inclusion of a forced number at the third character position prevents spelling ‘undesirable words’. A letter in the second character position allows for clear distinction between a code from ICD-11 and one from ICD-10. For further information refer to section [1.2.4.1Code structure](#).

3.2 Chapter Structure of ICD-11

The international core reference linearisation is the ICD-11 for Mortality and Morbidity Statistics (ICD-11MMS). It is used for coding and reporting illnesses or causes of death for international comparison. The naming of this linearisation highlights its two main use cases. This core linearisation is divided into 28 chapters, of which 25 refer to health conditions similar to past ICD versions, while one serves to identify external causes of morbidity and mortality, and another includes concepts of traditional medicine. Lastly, there are two additional sections for optional additional use, one for extension codes to add more detail for different dimensions of a disease, such as anatomy, mark a condition to be present on admission, or a disease having been relevant in the family history (see Section [Extension Codes]) and the other for functioning assessment to provide a set of codes for assessment and scoring in the ICD using ICF functioning domains of high explanatory power (see Section [3.2.27 Section V – Supplementary section for functioning assessment](#)).

ICD-11 has five new chapters. As a result, the numbering of the chapters has changed. The new chapters are:

- Chapter 03 'Diseases of the blood or blood-forming organs' and Chapter 04 'Diseases of the immune system'. Conditions affecting the immune system' and conditions affecting the blood are now in two separate chapters.
- Chapter 07 'Sleep-Wake disorders'. Sleep wake disorders have been regrouped together in this new chapter.
- Chapter 17 'Conditions related to sexual health'. Sexual conditions have been grouped together in this new chapter.
- Chapter 26 'Supplementary Chapter Traditional Medicine Conditions - Module I'. A chapter for traditional medicine has been added.

The following is an overview of the organisational principles and classification structure (hierarchy) for each of the 26 chapters. The revised structure and new sets of functionalities in ICD-11 were the result of incorporating scientific updates and making the classifications more relevant for computerisation.

3.2.1 Chapter 01 – Certain infectious or parasitic diseases

Structure of Chapter 01

3.2.1.1 Chapter 01 – Structure of chapter 01

The chapter first lists infectious disease groupings by clinical syndromes, then groups other infections by mode of transmission, and then lists the remaining infections by their agent. Some conditions of major public health concern are listed at the same level. Variants to the conditions in the chapter that occasionally can occur as localised infections are primarily coded to this chapter. Infections that are primarily localised, and where the agent usually is either unknown, or not relevant, or there is a mixed aetiology, reside in the organ chapters. Frequent infectious agents may be listed as individual child categories under the localised infection. In some instances, infections could equally be located in the infectious disease chapter and in an organ system chapter. In such cases, the decision that creates the least change (ICD-10 legacy) has been chosen. Some grouping is also designed to be able to code

frequently reported imprecise information, as is the case for meningitis and encephalitis and respiratory infections.

A special tabulation list groups the infections by agents and is intended for special tabulation and reporting only.

3.2.1.2 Chapter 01 – Rationale for chapter 01

The purpose of the structure of chapter 01 is to minimise the impact on longitudinal statistics of major infections, to allow reporting of main infections that have a common lead symptom (e.g. diarrhoea) without mention of a specific agent. Influenza, though visibly affecting the respiratory tract, affects multiple parts of the body and is also of important public health concern. For that reason, it has been moved into the infectious chapter. Prion diseases can be transmissible, genetic or arise spontaneously. They are rare conditions that only affect the nervous system. Many are inherited. The presence of a specific gene is a prerequisite to developing a prion disease. In view of these facts, it was decided to keep the prion diseases grouped together and move the whole group to the neurology chapter.

3.2.1.3 Antimicrobial resistance

The ICD parts relating to Antimicrobial Resistance (AMR) have been designed to support the Global Antimicrobial Resistance Surveillance System. Priority pathogens are identified in combinations with currently relevant antimicrobial substances. The section is designed to allow postcoordination of other substance and agent combinations in a cluster. The section on AMR is located in Chapter 21 'Symptoms, signs or clinical findings, not elsewhere classified', so that the underlying disease or agent is always coded in conjunction with the AMR category. ICD and the surveillance system focus on specific tracer pathogen-substance combinations. However, ICD design allows the coding of the full antibiotic susceptibility pattern if desired. For tabulation, the AMR codes should be reported in combination with the infectious disease. Where only one condition can be reported, the infectious disease should be retained. However, at the national level, the set of infectious diseases and the number of AMR cases among the infection cases should be tabulated.

3.2.2 Chapter 02 – Neoplasms

3.2.2.1 Chapter 02 – Structure of chapter 02

The general hierarchy of Chapter 02 consists of the following:

- 1st level - Behaviour
- 2nd level - Broad sites or systems
- 3rd level - Specific site
- 4th level - Morphological (histological) type

There are three groups that are an exception to the above hierarchy. They are:

1. Neoplasms of brain and central nervous system
 - 1st level - Broad sites
 - 2nd level - Behaviour - morphological (histological) type
2. Neoplasms of haematopoietic and lymphoid tissues

1st level - Broad morphological (histological) type

2nd level - Specific morphological (histological) type

3. Malignant mesenchymal neoplasms

1st level - Specific morphological (histological) type

2nd level - Site

3.2.2.2 Chapter 02 - Rationale for Chapter 02

The progress in oncology has clearly demonstrated that a site-only based categorisation of malignant and benign tumours provides limited information for prevention, treatment, and prognosis for persons that are affected by a tumour. ICD–10 included a limited number of categories based on histopathology (e.g. some lymphoid neoplasms, melanoma).

In ICD–11, main tumour sites have subdivisions of histopathology first. The groups chosen were based on an analysis of international mortality and morbidity reporting, cancer registries, and clinical reporting. The redesigned sections were reviewed for missing details in relation to the ICD use cases.

Keeping the main anatomical axes intact allows backwards compatibility. However, the structure was adjusted in a few places to match anatomical subdivisions of the TNM classification (<https://www.uicc.org/resources/tnm>).

For tumours of the central nervous system, the histological and behavioural distinction between benign and malignant is a grey area. As such, it was decided to move all central nervous system tumours outside the basic framework of behaviour and group them together.

The field of genetic markers is rapidly changing. Whereas for some tumours, such markers have been used for many years, for others, this is not the case. As such, with the exception of haematological tumours, genetic markers were not included, and have not been used for the classification. They are, however, included in Chapter X ‘Extension codes’, and can be added as postcoordination to the relevant code from the neoplasms chapter to describe the relevant tumour entity fully.

3.2.3 Chapter 03 – Diseases of the blood or blood-forming organs

3.2.3.1 Chapter 03 – Structure of chapter 03

This new chapter (previously part of Chapter III in ICD-10) has three main sections:

- Anaemias or other erythrocyte disorders
- Coagulation defects, purpura or other haemorrhagic or related conditions
- Diseases of spleen

Neoplasms of haematopoietic and lymphoid tissues are primarily located in Chapter 02 ‘Neoplasms’ while Symptoms, signs or clinical findings of blood or blood-forming organs or the immune system are primarily located in Chapter 21.

The first two major sections comprise of the following hierarchy:

- 1st level - Anaemias and coagulation disorders
- 2nd level - Broad category of disease/disorder type
- 3rd level - Congenital vs acquired
- 4th level - Further specificity of disease/disorder type

The third major section comprises of the following hierarchy:

- 1st level - Diseases of spleen
- 2nd level - Congenital vs acquired
- 3rd level - Specific disease/disorder type

3.2.3.2 Chapter 03 – Rationale for chapter 03

For Chapter 03, there has been a reorganisation of the chapter into a clinical view of diseases of the blood, an aetiological view of diseases of the blood and diseases of the spleen. Anaemias are now all under one group with a separate group for 'Coagulation defects, purpura or other haemorrhagic correlated conditions'.

3.2.4 Chapter 04 – Diseases of the immune system

3.2.4.1 Chapter 04 – Structure of chapter 04

This new chapter (previously part of Chapter III in ICD-10) has the following sections:

Immunodeficiencies

Non-organic specific systemic disorders

- 1st level - Being the main groupings above
- 2nd level - Broad category of disease/disorder type
- 3rd level - Specific disease/disorder type
- 4th level - Further specificity of disease/disorder type

- 1st level - Autoinflammatory disorders
- 2nd level - Specific syndrome

- 1st level - Allergic or hypersensitivity conditions
- 2nd level - Broad category for body systems

- 1st level - Certain diseases involving the immune system
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

Diseases of thymus

- 2nd level - Specific disease/disorder type

3.2.4.2 Chapter 04 – Rationale for chapter 04

For Chapter 04, there are new sections for immune disorders that differ from the section previously located in Chapter III of ICD-10. For the immune system they are classified mainly by clinical syndrome, and in an alternate view the immune system conditions are shown by

cell line. A section for Allergic or hypersensitivity conditions has been included in this chapter. Overall, more detail has been added to the chapter.

3.2.5 Chapter 05 – Endocrine, nutritional or metabolic diseases

3.2.5.1 Chapter 05 – Structure of Chapter 05

Chapter 05 has four major sections:

1. Endocrine diseases
 - 2nd level - Specific gland or hormone system
 - 3rd level - Specific diseases/disorder
2. Nutritional disorders
 - 2nd level - Broad categories of diseases/disorder
 - 3rd level - Specific disease/disorder
3. Metabolic disorders
 - 2nd level – Broad categories of diseases/disorder
 - 3rd level - Specific disease/disorder
4. Postprocedural endocrine or metabolic disorders
 - 2nd level - Specific disease/disorder

Neoplasms of the endocrine system are primarily located in Chapter 02 ‘Neoplasms’ and Symptoms, signs or clinical findings of endocrine, nutritional or metabolic diseases are primarily located in Chapter 21.

3.2.5.2 Chapter 05 – Rationale for Chapter 05

There is increased international standardisation of endocrine disease terminology being used to describe the complex nature of endocrine conditions. The intent is to include all dysfunctions that lead to a specific endocrine disorder.

Diabetes mellitus and Intermediate hyperglycaemia have been expanded to reflect current international terminology. The complications often associated with diabetes have continued to be included in the classification in the appropriate body system chapter in line with the various clinical modifications. ‘Code also’ and ‘Use additional code’ notes have been included to link the types of diabetes and the various complications to enable the addition of codes for further specificity.

Sources of change for this section were based on the current WHO Classification of Diabetes Mellitus and Intermediate Hyperglycaemia 2011 and the Department of Chronic Diseases, Health Promotion, WHO.

The WHO Department of Nutrition for Health and Development proposed changes to the section on Nutritional Disorders with advice from the Nutrition Guidance Expert Advisory Group (NUGAG) for updates to this section of the classification. Metabolic disorders are now

aetiologically based and have been classified into three distinct areas; ‘Inborn errors of metabolism’, ‘Disorders of metabolite absorption and transport’ and ‘Disorders of fluid, electrolyte and acid-base balance’ following clinical advice received from the relevant international societies for metabolic diseases.

3.2.6 Chapter 06 – Mental, behavioural or neurodevelopmental disorders

3.2.6.1 Chapter 06 – Structure of Chapter 06

The hierarchy of Chapter 06 consists of:

- 1st level - Broad category of disease/disorder type
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

3.2.6.2 Chapter 06 – Rationale for Chapter 06

The overall linear structure of the proposed Mental, behavioural or neurodevelopmental disorders chapter for ICD–11 has been a topic of substantive and comprehensive discussions by the Topic Disorders Advisory Group for Mental Health, as well as extensive interactions with the American Psychiatric Association in relation to the just-published Fifth Edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) (1), from the time of the Advisory Group’s initial appointment in 2007.

The appropriate architecture of a diagnostic classification of mental and behavioural disorders is an issue that has received substantial attention over the course of the revision (e.g. 2-4). One of the guiding principles of the ICD–11 is that it should reflect current scientific evidence regarding the relationships among disorders (5) rather than antiquated concepts such as ‘neurosis’, which have poor construct and predictive validity. In addition, a major goal of the WHO Department of Mental Health and Substance Abuse for the current revision is to improve the clinical utility of this part of the ICD–11 (6, 7). Because the ICD–11 uses a different coding structure that is not based on a decimal numbering system, such that a larger number of blocks or groupings can be accommodated within the chapter, an important opportunity was presented to bring the classification more in line with current research and clinical practice in terms of how groupings of disorders are represented.

Three streams of work provide the rationale and evidence for the linear structure of Mental and Behavioural Disorders in the ICD–11.

Evidence Reviews by Working Groups for ICD–11 Mental, behavioural or neurodevelopmental disorders

The first stream of work relates to the outcome of evidence reviews by the 14 Working Groups reporting to the Advisory Group, each of which had multiple face-to-face meetings over at least a 2-year period. The Working Groups were asked to review the available scientific evidence and other information about the clinical application of classifications in various settings throughout the world, and to provide evidence and a rationale for its groupings as well as the content and arrangement of categories within them. This work resulted in manuscripts describing the rationale for most groupings of disorders that have been published in or submitted to peer-reviewed journals (e.g. 8-15). Space does not permit detailing the rationale and evidence base for each structural change here, but this

information as it relates to any specific decision can be provided on request based on the material generated by the Working Groups.

Formative Field Studies on Clinical Utility of the Linear Structure

The second stream of work relevant to the linear structure of Mental and Behavioural Disorders focused on clinical utility and is represented by two formative field studies undertaken by the WHO and the Field Studies Coordination Group reporting to the Advisory Group (16, 17). The purpose of these studies was to examine the conceptualizations held by mental health professionals around the world of the relationships among mental disorders to inform decisions about the structure of the classification. From a clinical utility perspective, particularly in terms of improving the interface between health information and clinical practice, the most important and desirable features of a classification's organisation is that (a) it helps clinicians find the categories that most accurately describe the patients they encounter as quickly, easily, and intuitively as possible and (b) the diagnostic categories so obtained would provide them with clinically useful information about treatment and management. A mental disorders classification that is difficult and cumbersome to implement in clinical practice and does not provide information that is of immediate value to the clinician has no hope of being implemented accurately at the encounter level in real-world health care settings. In that event, clinical practice will not be guided by the standardisation and operationalization of concepts and categories that are inherent in the classification, and important opportunities for practice improvement and outcomes assessment will be lost. In turn, a diagnostic system that is characterized by poor clinical utility at the encounter level cannot generate data based on those encounters that will be a valid basis for health programs and policies, or for global health statistics. The rationale behind these two studies was that if the ways in which clinicians conceptualized the organisation of mental disorders as encountered in their day-to-day clinical practice was found to be (a) consistent across countries, languages, and disciplines, and (b) distinct from the organisation of ICD-10, then this information could be used to create a classification of mental disorders that corresponds more closely to clinicians' cognitive organisation of categories and would therefore be more intuitive and efficient for use in real-world health care settings.

The first formative field study (17) was an internet-based study administered in both English and Spanish, in which 1,371 psychiatrists and psychologists from 64 countries participated. The second formative field study (16) involved the face-to-face administration of a standardised sorting and hierarchy-formation task to 517 mental health professionals in eight countries and five languages. Both studies found that clinicians' conceptual map of mental disorders was rational and highly stable across profession, language, and country income level. Moreover, both studies found that the proposed structure for mental and behavioural disorders in ICD-11 was more consistent with clinicians' conceptual models than the structure of either ICD-10 or DSM-IV. The second study also clearly demonstrated that clinicians preferred a 'flatter' structure with a larger number of groupings as compared with a more hierarchical structure with fewer groupings as found in ICD-10.

Harmonisation with DSM-5

The third stream of work relates to efforts to harmonise the structure of the ICD-11 chapter on Mental and Behavioural Disorders with the structure of the DSM-5, where possible.

Overall, the high degree of similarity between the overall structure of DSM-5 (1) and the proposed linear structure for ICD–11 Mental and Behavioural Disorders represents a major success of the ICD – DSM harmonisation effort. Relatively minor differences relate primarily to:

1. proposals to combine the classifications of ‘organic’ and ‘non-organic’ aspects of conditions such as sleep disorders and sexual dysfunctions in ICD–11 in separate chapters in ways that are more consistent with current evidence and clinical practice, which was not an option for DSM-5 given that it is by definition a classification of mental disorders; and
2. differences in conventions related to residual categories and mental disorders associated with other underlying disease under ICD–11 from decisions about the organisation of such categories in DSM-5. Additional information about the rationale for the few remaining substantive differences in overall structure between the two classifications is available upon request. It must be emphasized that the resulting similarity in organisation between the two systems is the product of several years of complex negotiations. Given that DSM-5 has already been published, further changes to the ICD–11 structure would almost certainly move ICD–11 in the direction of reduced similarity and harmonisation with DSM-5.

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3.2.7 Chapter 07 – Sleep-wake disorders

3.2.7.1 Chapter 07 – Structure of Chapter 07

Chapter 07 is a new chapter in ICD–11. It contains Sleep-wake disorders that were previously located within the respiratory, neurology, or mental health chapters. By combining these disorders into one chapter, more detail can be included for many of the sleep related disorders. The hierarchy consists of:

- 1st level - Broad category of disease/disorder type
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

3.2.7.2 Chapter 07 – Rationale for Chapter 07

As Sleep-wake disorders pertain to an area of overlap between mental health, neurological disorders and pulmonary conditions, the decision was made to place them together in one chapter.

3.2.8 Chapter 08 – Diseases of the nervous system

3.2.8.1 Chapter 08 – Structure of Chapter 08

- 1st level - Mixture of diseases, disorders and sites and combinations of both.
- 2nd level - Subcategory mixture of specific disease or disorder type and sometimes site.

3.2.8.2 Chapter 08 – Rationale for Chapter 08

ICD–11 sees a major overhaul in the organisation of the blocks which make up the neurology chapter. The restrictive decimal coding system of the ICD–10, with its capacity to contain only 11 blocks of disorders per chapter, resulted in blocks containing miscellaneous neurological entities which did not logically fit together, such as the episodic and paroxysmal disorders block, containing headache disorders, epilepsy, transient ischaemic attacks and sleep disorders. The ICD–11 now positions headache disorders, epilepsy and cerebrovascular disorders at a block level, and sleep disorders at chapter level (Chapter 07).

Not only has the structure of the neurological chapter changed, but the approach to classification also integrates current clinical practice and advancements in the understanding of neurological diseases. In the time since the ICD–10 was published, enormous progress in the fields of genetics, molecular biology and medical technologies have been made. An increase in the number of codes is inevitable when one reflects on the recent knowledge gain in neurology, so a balance between comprehensiveness, clinical utility and maintaining a public health approach is the aim. The working groups tackled this issue by considering the more common disorders to appear in the chapter, with less common aetiological variations of these disorders being subject to a ‘double coding’ technique. One major change which illustrates the advancement of knowledge is the addition of a block entitled ‘Paraneoplastic and autoimmune disorders of the nervous system’. This block contains immune-mediated neurological diseases, a field in which

knowledge has exploded in recent years. A second example of how the new version reflects molecular biological advancement is through awarding Prion diseases block status despite their rarity. Previously, they featured as part of the infections of the central nervous system block, but research interest after the major public health issue in Europe in the 1990s has led to new variants of prion diseases being discovered.

The world has seen a large rise in the elderly population since the 1990s. Neurocognitive disorders have been declared as a major public health concern and research into its aetiology and neuropharmacology has boomed. The ICD–11 block on Neurocognitive disorders reflects the better understanding in this area.

One final particularly noteworthy change can be found in the ‘Other disorders of the nervous system’ block. This block is employed to capture the ‘spill over’ from other neurology blocks and those disorders which are deemed unclassifiable elsewhere. In the ICD–10, due to the aforementioned decimal coding system, this block was an incongruent collection of diseases. This block has now reduced significantly in size due to the new, streamlined neurology chapter structure which includes new blocks of disorders previously contained in the ‘other disorders of the nervous system’ section of ICD–10. These include ‘disorders of consciousness’, ‘disorders of cerebrospinal fluid pressure and flow’, ‘disorders of the autonomic nervous system’, ‘nutritional and toxic disorders of the nervous system’ and ‘spinal cord disorders excluding trauma’. Their promotion to block status will hopefully have a positive effect on coding practices.

One complicating issue facing the Neurology Topic Advisory has been the need to cross-link disorders which have a neurological presentation or phenotype to their aetiological roots within other chapters or blocks within the neurology chapter. One of the countless examples of this kind of relationship would be mitochondrial disorders of neuromuscular junction. They must be cross-linked both in the neurology chapter, and in the Endocrine, nutritional or metabolic diseases chapter.

3.2.9 Chapter 09 – Diseases of the visual system

3.2.9.1 Chapter 09 – Structure of Chapter 09

The general hierarchy of Chapter 09 consists of the following:

- 1st level - Broad category of anatomy
- 2nd level - Specific anatomy category
- 3rd level - Broad category of disease/disorder type
- 4th level - Further specificity of disease/disorder type

3.2.10 Chapter 10 - Diseases of the ear or mastoid process

3.2.10.1 Chapter 10 – Structure of Chapter 10

The general hierarchy of Chapter 10 consists of the following:

- 1st level - Broad category of anatomy
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

3.2.11 Chapter 11 – Diseases of the circulatory system

3.2.11.1 Chapter 11 – Structure of Chapter 11

There are two main hierarchies in Chapter 11.

- 1st level - Broad category of disease/disorder type
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

OR

- 1st level - Broad category of anatomy
- 2nd level - Specific anatomy type
- 3rd level - Specific disease/disorder type

3.2.11.2 Chapter 11 – Rationale for Chapter 11

There have been large scale changes in clinical practice in cardiovascular diseases and their management since ICD-10 was published over 20 years ago. Changes introduced for ICD-11 in this chapter reflect these changes and the shift in disease profiles and increased survival following procedures. As a consequence, there has been a major expansion in the number of disease entities within ICD-11, with new classification hierarchies and updated nomenclature. For instance, the incidence of heart valve disease is no longer dominated by rheumatic fever in developed societies, although it remains important in developing nations, and consequently there has been a shift in diagnostic paradigms to that of valve type, then valve pathology followed by aetiology.

Many items previously classified in ICD-10 as 'Other forms of heart disease' (I30-I52) have become major clinical issues in today's cardiology, warranting the creation of new distinct higher-level categories. Two examples are:

- Diseases of the myocardium, including extensive subsections on Myocarditis and Cardiomyopathy.
- Cardiac arrhythmia, including a large new subsection on 'Cardiac arrhythmia associated with genetic disorder' and 'Pacemaker or implantable cardioverter or defibrillator or lead dysfunction', both of which are increasingly important areas of clinical practice. The changes in this section have had major input and endorsement from the Paediatric & Congenital Electrophysiology Society and the International Society for Nomenclature of Paediatric and Congenital Heart Disease.

The change in the ICD revision process to be clinically driven has meant that areas primarily managed by non-cardiologists have been relocated to more suitable chapters. Thus, Cerebrovascular diseases have been reclassified to Chapter 08, 'Diseases of the nervous system' and oesophageal varices have been relocated to Diseases of the digestive system (Chapter 13).

A new subsection on Pulmonary Hypertension in the Pulmonary heart disease and diseases of pulmonary circulation section, is based on the resulting paper Updated Clinical Classification of Pulmonary Hypertension, following the 5th World Symposium held in Nice, France, in 2013.

The postprocedural disorders section has been markedly enlarged reflecting increased survival after cardiovascular procedures over the last two decades with recognition of an increasing number of patients with postprocedural morbidities and disease specific complications.

The section on Congenital anomaly of heart and great vessels and related acquired abnormalities classified to Chapter 20 'Developmental anomalies' has been based on the International Paediatric and Congenital Cardiac Code (IPCCC), which has been created over the last decade by the International Society for Nomenclature of Paediatric and Congenital Heart Disease (ISNPCHD, <http://www.ipccc.net>). As a consequence, the 73 congenital cardiology ICD-10 entities in Q20-Q29 have been expanded to 316 diagnoses, as an accurate summation of the heterogeneity of cardiac malformations seen in clinical practice. Reference was also made to the Anatomic and clinical classification of congenital heart defects (ACC-CHD) with the corresponding IPCCC and ICD-10 codes.

3.2.12 Chapter 12 – Diseases of the respiratory system

3.2.12.1 Chapter 12 – Structure of Chapter 12

There are two main hierarchies in Chapter 12:

- 1st level - Broad category of disease/disorder type
- 2nd level - Specific disease/disorder type with some anatomy included
- 3rd level - Further specificity of disease/disorder type

OR

- 1st level - Broad category of anatomy
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

3.2.12.2 Chapter 12 – Rationale for Chapter 12

The changes to Chapter 12 have been made principally to provide current clinical terminology and classification of conditions primarily affecting the respiratory system and have been based on input from international societies and stakeholders. Infectious lung diseases have been moved to Chapter 01 to better reflect the infectious nature of these conditions. Neoplasms of the respiratory system are in Chapter 02 'Neoplasms', and Developmental respiratory diseases are now located in Chapter 20 'Developmental anomalies'.

The grouping 'Upper respiratory tract disorders' contains upper respiratory tract diseases except for conditions that moved to the Infectious disease chapter.

The Lower respiratory tract diseases shifted from the Chronic lower respiratory diseases of the ICD-10, but Chronic obstructive pulmonary disease (COPD) was made an independent category based on an international concept.

Cystic fibrosis has been moved to Certain lower respiratory tract diseases and multiparented to metabolic disorders in the Endocrine chapter because: 'The representative clinical conditions of cystic fibrosis are intractable respiratory infection, end stage

respiratory failure, exocrine pancreatic insufficiency and digestive organ lesions such as the meconium ileus. Cystic fibrosis is a disease due to an abnormality of the Cl ion channel which is CFTR, symptoms of the respiratory symptom is recognised in nearly all cases of patients. The cause of death is mainly respiratory abnormality, and this disease is the target disease of lung transplantation.' This description of cystic fibrosis is found in representative textbooks ('Diseases of the Airways' in the textbook 'Fraser and Pare's Disease of the Chest').

- 'OBSTRUCTIVE DISEASES' in the textbook 'Murray and Nadel's Textbook of Respiratory Medicine'
- 'OBSTRUCTIVE LUNG DISEASES' in the textbook 'Fishman's pulmonary diseases and disorders'
- 'Disease of the Airways' in the textbook 'Fraser and Pare's Disease of the Chest' 'Pulmonary Diseases' in the textbook 'Washington Manual of Medical Therapeutics, The, 34ed.'

The section pertaining to Inhalation, occupational and environmental lung disease has been based on input from the WHO Occupational Health Division.

The Certain specified respiratory diseases principally affecting the lung interstitium shifted from the Other respiratory diseases principally affecting the interstitium. The Idiopathic interstitial pneumonitis was made an independent category based on an international concept and the category of the Primary interstitial lung diseases specific to infancy and childhood was created independently based on the proposal of the Paediatric Topic Advisory Group (TAG).

The section of the Certain diseases of the respiratory system and the section of the Postprocedural respiratory disorders were shifted from Other diseases of the respiratory system of ICD-10 except for the Mediastinal and diaphragm disorders that moved to the section of Pleural, diaphragm and mediastinal disorders.

3.2.13 Chapter 13 – Diseases of the digestive system

3.2.13.1 Chapter 13 – Structure of Chapter 13

The general hierarchy of Chapter 13 consists of the following:

- 1st level - Detailed anatomy
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

3.2.13.2 Chapter 13 – Rationale for Chapter 13

ICD-11 has been improved in structure and content to include diseases and disorders of the orofacial complex. There are several other tissues which as essential components of the orofacial complex, have an important function, and their impairment will have a direct impact on oral health status. It is important to recognise that oral health is more than having healthy teeth; having oral health is being free of chronic oral-facial pain conditions, oral and pharyngeal cancers, oral soft tissue lesions, periodontal (gum) disease, tooth decay and tooth loss and tooth surface loss, birth defects such as cleft lip and palate, and scores of other diseases and disorders that affect the oral, dental, and craniofacial tissues (orofacial

complex) as well as associations with systemic health and disease. This underlines the importance of providing a coherent system for coding and classifying data on orofacial complex diseases and disorders so that the oral health professional can record and collect data from each patient at their clinics, regardless of whether such facility may be part of large hospitals, or small clinics. It is anticipated that being able to record and interpret such data will enable health professionals to contribute to the improvement of oral health as an essential component of general health and will stimulate the use of ICD-11 by oral health personnel.

Major changes have been made to this chapter with very detailed anatomical groups being added to the hierarchy for the digestive tract, according to rostral-caudal order, with the exception of categories for hernia, functional gastrointestinal disorders, and inflammatory bowel diseases.

Functional gastrointestinal disorders are independently described because their pathophysiology is considered from the standpoint of 'Brain-Gut axis', and not only from their impact on the gastrointestinal tract. Inflammatory bowel diseases are also independently described mainly because Crohn's disease involves several organs. In each anatomical group (organ group), aetiology based classifications are used to sub-classify disorders. Particularly, GI disorders are arranged in the following categories:

- A. Acquired anatomical or morphological alterations
- B. Motor disorders
- C. Inflammation including ulcer
- D. Vascular disorders
- E. Non-neoplastic polyps

In addition, there are two other categories listed, although Chapter 13 is not the primary place for these disorders.

F. Structural developmental anomalies (located in Chapter 20 Developmental anomalies)

G. Neoplasms (located in Chapter 02 Neoplasms)

Important or common digestive diseases have been allocated their own category, for example gastro-oesophageal reflux disease, columnar metaplastic epithelium, intestinal malabsorption and protein-losing enteropathy, ulcerative colitis, non-alcoholic fatty liver disease and diverticular disease. Polyps are now classified independently, and not in the 'other diseases' section of anatomical site.

Common digestive diseases extending over several organs are classified principally into the disease category of the rostral organ. For example, 'Gastroenteritis' is classified in 'Gastritis', and 'Gastroduodenal ulcer' is classified in 'Gastric ulcer'. The item 'Peptic ulcer, site unspecified' should not be used due to advances in medical technology. It should be classified into either the 'Oesophageal ulcer, Gastric ulcer, Duodenal ulcer or Anastomotic ulcer' category, depending on the disease site.

Vascular disorders of GI organs have been allocated their own category. Oesophageal varices, gastric varices and haemorrhoids are now classified in Chapter 13. In 'Diseases of liver', there are new independent categories including Metabolic and transporter liver disease, Autoimmune liver diseases, Non-alcoholic fatty liver disease and Vascular disorders of the liver.

For the classification of Chronic liver disease with cirrhosis, 'Liver cirrhosis', an item in 'Hepatic fibrosis and cirrhosis', is used. For example, 'Chronic hepatitis B' and 'Liver cirrhosis', 'Chronic hepatitis C' and 'Liver cirrhosis', 'Autoimmune hepatitis' and 'Liver cirrhosis', 'Primary biliary cholangiopathy' and 'Liver cirrhosis', etc. There are new independent sections for 'Diseases of gallbladder and biliary duct' and 'Diseases of pancreas'. Within these new sections, there are new independent categories including Structural developmental anomalies, Congenital anomalies, Acquired anatomical alterations, Cholangitis, Cystic diseases of the pancreas, Chronic pancreatitis and Autoimmune pancreatitis.

3.2.14 Chapter 14 – Diseases of the skin

3.2.14.1 Chapter 14 – Structure of Chapter 14

The general hierarchy of Chapter 14 consists of the following:

- 1st level - Broad category of disease/disorder type
- 2nd level - Specific disease/disorder type with some anatomical site
- 3rd level - Further specificity of disease/disorder type

3.2.14.2 Chapter 14 – Rationale for Chapter 14

Major changes have been made to this chapter adding detail coming from the fusion of the American, British, and German dermatological terminologies.

3.2.15 Chapter 15 – Diseases of the musculoskeletal system or connective tissue

3.2.15.1 Chapter 15 – Structure of Chapter 15

The general hierarchy of Chapter 15 consists of the following:

- 1st level - Broad category of disease/disorder type
- 2nd level - Specific disease/disorder type with some anatomical site
- 3rd level - Further specificity of disease/disorder type

3.2.15.2 Chapter 15 – Rationale for Chapter 15

The American College of Rheumatology and European League Against Rheumatism (ACR/EULAR) Diagnostic Criteria for Rheumatoid Arthritis (under development) was used to inform the code hierarchy and content model attributes for Rheumatoid arthritis. Current literature informed the change of title of 'systemic connective tissue disorders' to 'non-organ specific systemic autoimmune disorders'. The changes to vasculitis were based on the classification of the Chapel Hill International Consensus Conference on the Nomenclature of Systemic Vasculitis.

The category 'Dermatopolymyositis' was changed to 'Idiopathic inflammatory myopathies' with a change of axes and introduction of further granularity.

The revisions to the classification of spondyloarthritis reflect current expert opinion with comments from Dr Robert Landewé, with a separation of axial and peripheral. Together, the axial and peripheral spondyloarthritis criteria cover the entire spectrum of what was formerly called (undifferentiated) spondyloarthritis and (ankylosing) spondylitis. There is re-arrangement of infective spondyloarthritis, with a secondary axis for the major types of infective process, i.e. bacterial, fungal etc., and supplementary codes to be used for the specific infection.

The new category for Auto-inflammatory syndromes is based on the work of the International Society of Systemic Auto-inflammatory Disease (ISSAID).

3.2.16 Chapter 16 – Diseases of the genitourinary system

3.2.16.1 Chapter 16 – Structure of Chapter 16

Chapter 16 has specific sections for Diseases of the female genitourinary system, Diseases of the male genitourinary system, Disorders of breast, Diseases of the urinary system and Postprocedural disorders of the genitourinary system.

The general hierarchy of Chapter 16 consists of the following:

- 1st level - Broad category of body system
- 2nd level - Broad disease/disorder type (with some anatomy)
- 3rd level - Specific disease/disorder type (with some anatomy)

3.2.16.2 Chapter 16 – Rationale for chapter 16

The changes to Chapter 16 are aimed at increasing the clinical utility of the classification by providing a more user-friendly hierarchical structure, increased international comparability and standardisation of genitourinary conditions. This is accomplished by including the most scientifically accurate and internationally agreed-upon terms and definitions provided by various international stakeholders, including the WHO department of Reproductive Health and Research, the International Federation of Gynaecology and Obstetrics (FIGO), National Kidney Foundation and the Kidney Disease International Global Outcomes (KDIGO).

The chapter hierarchy is subdivided into Diseases of the Female Genital System, Diseases of the Male Genital System and Diseases of the Urinary system. This architecture of the female genital system and the male genital system was designed to improve the end-user experience. The female genital system hierarchy is broken down into non-inflammatory and inflammatory disorders, and then further divided by anatomical grouping in the order of gynaecologic (and obstetric) examination (from external to internal genitalia), where applicable. (Vulva, Vagina, Cervix, Uterus, Fallopian Tube, Ovary, Pelvic Cavity).

These groupings have further subdivisions for congenital and acquired abnormalities, as appropriate.

To reflect the current scientific understanding for certain genitourinary conditions, additional detail has been included for the following areas:

Amenorrhea

Ovarian dysfunction

Female pelvic pain

Endometriosis

Adenomyosis

Female infertility

Male infertility

Early pregnancy loss

Pregnancy outcomes

The Kidney failure section of the classification has been revised to reflect the current evidence-based definitions of acute kidney versus chronic kidney disease and the new Kidney Disease ('Improving Global Outcomes (KDIGO) definitions and staging system for acute kidney failure'.)

3.2.17 Chapter 17 – Conditions related to sexual health

3.2.17.1 Chapter 17 – Structure of Chapter 17

New chapter in ICD-11 divided into major sections for:

Sexual dysfunctions

Sexual pain disorders

Gender incongruence

1st level - Broad category of condition

2nd level - Specific type of condition

3rd level - Specific disease/disorder

3.2.17.2 Chapter 17 – Rationale for Chapter 17

The chapter has been formulated to group sexually related conditions. This also allows categorisation of gender identity related conditions without stigmatisation, while maintaining recognition of these entities as real conditions so that related health interventions can be accommodated within the health system.

3.2.18 Chapter 18 – Pregnancy, childbirth or the puerperium

3.2.18.1 Chapter 18 – Structure of Chapter 18

The general hierarchy of Chapter 18 consists of the following:

1st level - Broad category related to the stages of pregnancy, child birth or the puerperium

2nd level - Specific disease/disorder type

3rd level - Further specificity of disease/disorder type

3.2.18.2 Chapter 18 – Rationale for Chapter 18

The changes to this chapter are intended to increase the clinical utility of the classification by providing a more user-friendly hierarchical structure. Increasing the international comparability and standardisation of conditions related to pregnancy, childbirth and the puerperium by including the most scientifically accurate and internationally agreed-upon terms and definitions provided by various international stakeholders, such as the WHO department of RHR, International Federation of Gynaecology and Obstetrics (FIGO), was also a highly important aspect of the modifications. Particular attention was given to correct integration of concepts and definitions of the International Committee Monitoring Assisted Reproductive Technologies (ICMART).

The changes reflect the current understanding for certain conditions related to pregnancy, childbirth and the puerperium. Additional specifications have been included for Early pregnancy loss

3.2.19 Chapter 19 – Certain conditions originating in the perinatal period

3.2.19.1 Chapter 19 – Structure of Chapter 19

The general hierarchy of Chapter 19 consists of the following:

- 1st level - Broad category disease/disorder type and some anatomy
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

3.2.20 Chapter 20 – Developmental anomalies

3.2.20.1 Chapter 20 – Structure of Chapter 20

Chapter 20 has undergone a major restructure with it now having four major sections

Structural developmental anomalies primarily affecting one body system

- 1st level - Broad category of anatomy
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

Multiple developmental anomalies or syndromes

- 1st level - Broad category of anatomy
- 2nd level - Specific disease/disorder type
- 3rd level - Further specificity of disease/disorder type

Chromosomal anomalies, excluding gene mutations

- 1st level - Specific disease/disorder type
- 2nd level - Further specificity of disease/disorder type

Conditions with disorders of intellectual development as a relevant clinical feature

- 1st level - Non-syndromic vs syndromic conditions
- 2nd level - Further specificity of disease/disorder type

3.2.20.2 Chapter 20 – Rationale for Chapter 20

The ICD–10 classification of developmental anomalies is covered by Chapter XVII: Q00-Q99 Congenital malformations, deformations and chromosomal abnormalities.

It is a very heterogeneous chapter, including malformations, genetic syndromes (with or without malformations) and chromosomal anomalies. This leads to confusion between the genetic origin of a disease and malformation. Therefore, all genetic syndromes without structural developmental anomalies are excluded from this chapter and are reallocated to appropriate chapters of the ICD-11, according to the affected body system(s).

The new chapter 20 has three main divisions:

- Structural developmental anomalies/malformations
- Multiple developmental anomalies and syndromes
- Chromosomal anomalies and genetic defects

The first division ‘Structural developmental anomalies/malformations’ includes isolated conditions affecting only one body system. It is organised in sections corresponding to those body systems, which are also classified in the other relevant chapters of ICD–11.

The second division ‘Multiple developmental anomalies and syndromes’ includes conditions affecting several locations within one body system, or several body systems simultaneously. Syndromes which can be said to affect one body system predominantly are assigned to corresponding sections within this division. Syndromes which affect several body systems, without one clearly predominating, are put together in a specific section at the end of the division. There is also a section for Dysplasia syndromes due to inborn errors of metabolism, all of them primarily classified in the chapter for metabolic diseases.

The third division ‘Chromosomal anomalies and genetic defects’ departs from the clinical approach generally followed in the ICD and classifies developmental anomalies defined genetically or cytogenetically, since there is no clear-cut distinction between genetics and cytogenetics. We have started to include specific deletions and duplications corresponding to a clear phenotype, knowing that many more will be described in the coming years. Future ones will be added whenever necessary, during the post-publication revisions of the ICD–11.

A special problem is how to deal with diseases historically defined clinically but including a chromosomal/genetic anomaly as aetiology. In some cases, there are several aetiologies for the clinical entity, and not all of them are chromosomal anomalies: for instance, Silver-Russell syndrome can be caused by a 11p15 duplication, a 7p11.2p13 duplication, but also by maternal uniparental disomy of chromosome 7 or 11 and imprinting defects of 11p15. In other cases, there is an overwhelming correspondence between the clinical entity and a cytogenetic aetiology: for instance, Williams-Beuren syndrome corresponds to the 7q11.23 deletion.

Polyhierarchy is used in a restricted way within the frame of this chapter: once a disease is assigned to a section, it is generally not secondarily classified elsewhere in the chapter. The structure would otherwise become too intricate. On the other hand, all entities in this chapter are to be classified in other chapters of ICD–11, when appropriate.

3.2.21 Chapter 21 – Symptoms, signs or clinical findings, not elsewhere classified

3.2.21.1 Chapter 21 – Structure of Chapter 21

Chapter 21 is divided into major sections based on body systems. Each of these sections has the following categories, as appropriate:

- Symptoms and signs
- Clinical findings

An additional section is located at the end of this chapter for ill-defined and unknown causes of mortality.

3.2.21.2 Chapter 21 – Rationale for Chapter 21

The different chapters of ICD–10 included several clinical manifestation categories, some of them as asterisk codes. In order to simplify the structure, improve the use of postcoordination, and also to remove ‘ill-defined’ conditions from organ chapters, several former asterisk codes, additional detail for diverse conditions, and the said ill-defined conditions have been moved here. All follow the main organisation by anatomy, and the anatomical groupings have a secondary parent to the relevant organ chapter, improving the user guidance.

3.2.22 Chapter 22 – Injury, poisoning or certain other consequences of external causes

3.2.22.1 Chapter 22 – Structure of Chapter 22

The general hierarchy of Chapter 22 consists of the following:

1st level - Broad category of anatomy (e.g. head; hip & thigh)
2nd level - Broad category of injury type (e.g. fracture; open wound)
3rd level - Further specification

OR

1st level - Broad category of cause of injury
2nd level - Specific injury type
3rd level - Further specificity of injury type

3.2.22.2 Chapter 22 – Rationale for Chapter 22

The principles of the revision were:

- Maintain good back-compatibility with ICD–10, particularly by minimising change at the former three-character level. Change at the former four-character level is more extensive but has also been done with this principle in mind.
- Take account of the extensions to this chapter in clinical modifications of ICD–10 because: